SIX EQUIVALENT CHARACTERIZATIONS OF COMBINED FIBRE FINITENESS OVER A POINT

THE STACKS PROJECT AUTHORS

Editorial note. The source marks (4) implies (5) 'immediate' and (6) implies (5) 'clear'; both are single weakenings of the same statement form, carried as explicit obligations.

Allowed standard prerequisites (editorial).

- Weak characterization. lemma-weak-UR-finite-above-x (carded L26699). Gives (1) iff (3).
- U-finiteness characterization. lemma-U-finite-above-x (carded L26697). Used in (2) implies (1).
- R-finiteness characterization. lemma-R-finite-above-x (carded L26698). Used in (2) implies (1).
- Quasi-compactness is etale-local on the target. Morphisms of Spaces, lemma quasi-compact-local (cross-chapter, named). Gives (3) implies (4).
- Factorization of field points in the same class. Properties of Spaces, lemma equivalence-class-point-monomorphism (cross-chapter, named). Gives (4) implies (6).

Source reference. Original lemma 03JU: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 534–534. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 03JS: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 485–485. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 040U: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 687–687. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original remark 03II: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 459–459. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:decent-spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 03DP: focused-seeds/geometry/S9-decent-ur-finite-above-x/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03HG: focused-seeds/geometry/S9-decent-ur-finite-above-x/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03KG: focused-seeds/geometry/S9-decent-ur-finite-above-x/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0H2X: focused-seeds/geometry/S9-decent-ur-finite-above-x/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03H4: focused-seeds/geometry/S9-decent-ur-finite-above-x/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let X be an algebraic space over S. Let $x \in |X|$. The following are equivalent:*

1. *for every affine scheme U, any étale morphism $\varphi : U \to X$ setting $R = U \times_X U$ the fibres of both*
$$|U| \longrightarrow |X| \quad \textit{and} \quad |R| \longrightarrow |X|$$
over x are finite,
2. *there exist schemes U_i and étale morphisms $U_i \to X$ such that $\coprod U_i \to X$ is surjective and for each i, setting $R_i = U_i \times_X U_i$ the fibres of both*
$$|U_i| \longrightarrow |X| \quad \textit{and} \quad |R_i| \longrightarrow |X|$$
over x are finite,
3. *there exists a monomorphism $\mathrm{Spec}(k) \to X$ with k a field in the equivalence class of x, and for any affine scheme U and étale morphism $U \to X$ the fibre product $\mathrm{Spec}(k) \times_X U$ is a finite scheme over k,*
4. *there exists a quasi-compact monomorphism $\mathrm{Spec}(k) \to X$ with k a field in the equivalence class of x,*
5. *there exists a quasi-compact morphism $\mathrm{Spec}(k) \to X$ with k a field in the equivalence class of x, and*
6. *every morphism $\mathrm{Spec}(k) \to X$ with k a field in the equivalence class of x is quasi-compact.*

Proof. The equivalence of (1) and (3) follows on applying Lemma 040U to every étale morphism $U \to X$ with U affine. It is clear that (3) implies (2). Assume $U_i \to X$ and R_i are as in (2). We conclude from Lemma 03JS that for any affine scheme U and étale morphism $U \to X$ the fibre of $|U| \to |X|$ over x is finite. Say this fibre is $\{u_1, \ldots, u_n\}$. Then, as Lemma 03JU (1) applies to $U_i \to X$ for some i such that x is in the image of $|U_i| \to |X|$, we see that the fibre of $|R = U \times_X U| \to |U| \times_{|X|} |U|$ is finite over (u_a, u_b), $a, b \in \{1, \ldots, n\}$. Hence the fibre of $|R| \to |X|$ over x is finite. In this way we see that (1) holds. At this point we know that (1), (2), and (3) are equivalent.

If (4) holds, then for any affine scheme U and étale morphism $U \to X$ the scheme $\mathrm{Spec}(k) \times_X U$ is on the one hand étale over k (hence a disjoint union of spectra of finite separable extensions of k by Remark 03II) and on the other hand quasi-compact over U (hence quasi-compact). Thus we see that (3) holds. Conversely, if $U_i \to X$ is as in (2) and $\mathrm{Spec}(k) \to X$ is a monomorphism as in (3), then

$$\coprod \mathrm{Spec}(k) \times_X U_i \longrightarrow \coprod U_i$$

is quasi-compact (because over each U_i we see that $\mathrm{Spec}(k) \times_X U_i$ is a finite disjoint union spectra of fields). Thus $\mathrm{Spec}(k) \to X$ is quasi-compact by Morphisms of Spaces, Lemma 03KG.

It is immediate that (4) implies (5). Conversely, let $\mathrm{Spec}(k) \to X$ be a quasi-compact morphism in the equivalence class of x. Let $U \to X$ be an étale morphism with U affine. Consider the fibre product

$$\begin{array}{ccc} F & \longrightarrow & U \\ \downarrow & & \downarrow \\ \mathrm{Spec}(k) & \longrightarrow & X \end{array}$$

Then $F \to U$ is quasi-compact, hence F is quasi-compact. On the other hand, $F \to \mathrm{Spec}(k)$ is étale, hence F is a finite disjoint union of spectra of finite separable extensions of k (Remark 03II). Since the image of $|F| \to |U|$ is the fibre of $|U| \to |X|$ over x (Properties of Spaces, Lemma 03H4), we conclude that the fibre of $|U| \to |X|$ over x is finite. The scheme $F \times_{\mathrm{Spec}(k)} F$ is also a finite union of spectra of fields because it is also quasi-compact and étale over $\mathrm{Spec}(k)$. There is a monomorphism $F \times_X F \to F \times_{\mathrm{Spec}(k)} F$, hence $F \times_X F$ is a finite disjoint union of spectra of fields (Schemes, Lemma 03DP). Thus the image of $F \times_X F \to U \times_X U = R$ is finite. Since this image is the fibre of $|R| \to |X|$ over x by Properties of Spaces, Lemma 03H4 we conclude that (1) holds. At this point we know that (1) – (5) are equivalent.

It is clear that (6) implies (5). Conversely, assume $\mathrm{Spec}(k) \to X$ is as in (4) and let $\mathrm{Spec}(k') \to X$ be another morphism with k' a field in the equivalence class of x. By Properties of Spaces, Lemma 0H2X we have a factorization $\mathrm{Spec}(k') \to \mathrm{Spec}(k) \to X$ of the given morphism. This is a composition of quasi-compact morphisms and hence quasi-compact (Morphisms of Spaces, Lemma 03HG) as desired. □